

Loading a custom structure doesn't load rails correctly

[MC-104231](#) STRUCTURES

Status	Resolved / Fixed
Affected	Minecraft 1.10; Minecraft 1.10.1; Minecraft 1.10.2; Minecraft 16w33a; Minecraft 1.12.2; Minecraft 1.13-pre3; 1.16.1; 20w28a; 1.16.3; 1.17.1; 1.18; 1.19; 1.20.1; 1.20.4; 1.21
Fixed in	25w16a
Reported	2016-06-22
Mojira category	Structures

REPRODUCTION TARGET

Version: Minecraft Java Edition 1.21 (vanilla)
Environment: Vanilla Minecraft Java Edition 1.21 in a single-player Superflat world using the integrated server. Use Creative mode with commands/cheats enabled. Do not enable experimental features, mods, data packs other than vanilla, or resource packs that obscure rail shapes.

PRECONDITIONS

- Minecraft Java Edition 1.21 is installed and can be selected as an exact launcher installation; 1.21 is explicitly listed as affected in the supplied report.
- The test world is disposable, since the procedure clears and rebuilds blocks near $X=0$, $Y=100$, $Z=0$ and $X=20$, $Y=100$, $Z=0$.
- Commands are enabled. If a world was not created with Allow Cheats enabled, use Esc -> Open to LAN -> Allow Cheats: ON -> Start LAN World before continuing.
- The player is in Creative mode and can fly; default controls are assumed, but rebound equivalents may be used.
- The original Step1.png and Step2.png attachments are unavailable in the supplied report. The plan therefore uses a deterministic, compact arrangement of two neighboring rail loops as a practical substitute for the pictured rail art.

STEPS TO REPRODUCE

1 REPORT

In the Minecraft Launcher, create or select an installation for the exact release Minecraft Java Edition 1.21, launch it, and confirm that no mods are loaded.

INPUT `Minecraft Java Edition 1.21`

SUCCESS CUE The 1.21 main menu is displayed and the client is identified as an unmodified Java Edition client.

2 INFERRED

Select Singleplayer -> Create New World. Set Game Mode to Creative, enable Allow Commands/Cheats, choose Superflat as the world type, leave the seed unspecified, and create the world.

SUCCESS CUE A new Superflat world opens with the player in Creative mode.

3 INFERRED

Wait until the world is ready. Do not continue while the loading overlay is present or nearby chunks are still appearing.

SUCCESS CUE Terrain is rendered and movement, flight, and chat commands respond normally.

4 INFERRED

Press T to open chat, enter the exact teleport command including the leading slash, and press Enter. This moves the player above and outside the construction regions.

INPUT `/tp @s 12 105 -8`

SUCCESS CUE The player is flying near X=12, Y=105, Z=-8 and can see the flat world below.

5 INFERRED

Open chat, execute the exact fill command, and press Enter. This clears both the source and destination volumes without clearing the player's position.

INPUT `/fill -5 95 -5 30 110 10 minecraft:air`

SUCCESS CUE The command reports that blocks were filled or changed, and the volume from the source area through the destination area is empty.

6 INFERRED

Open chat and execute the exact command to create a 7 by 7 stone support floor for the source rail structure.

INPUT `/fill 0 100 0 6 100 6 minecraft:stone`

SUCCESS CUE A 7 by 7 stone platform is visible from X=0 through X=6 and Z=0 through Z=6 at Y=100.

7 INFERRED

Open chat and execute the exact command to place the north straight section of the outer rail loop.

INPUT `/fill 1 101 0 5 101 0 minecraft:rail[shape=east_west]`

SUCCESS CUE Five rails occupy X=1 through X=5 at Y=101, Z=0.

8 **INFERRED**

Open chat and execute the exact command to place the south straight section of the outer rail loop.

INPUT `/fill 1 101 6 5 101 6 minecraft:rail[shape=east_west]`

SUCCESS CUE Five rails occupy X=1 through X=5 at Y=101, Z=6.

9 **INFERRED**

Open chat and execute the exact command to place the west straight section of the outer rail loop.

INPUT `/fill 0 101 1 0 101 5 minecraft:rail[shape=north_south]`

SUCCESS CUE Five rails occupy X=0 at Y=101, Z=1 through Z=5.

10 **INFERRED**

Open chat and execute the exact command to place the east straight section of the outer rail loop.

INPUT `/fill 6 101 1 6 101 5 minecraft:rail[shape=north_south]`

SUCCESS CUE Five rails occupy X=6 at Y=101, Z=1 through Z=5.

11 **INFERRED**

Open chat and execute the exact command for the northwest corner of the outer loop.

INPUT `/setblock 0 101 0 minecraft:rail[shape=south_east]`

SUCCESS CUE A rail is present at X=0, Y=101, Z=0.

12 **INFERRED**

Open chat and execute the exact command for the northeast corner of the outer loop.

INPUT `/setblock 6 101 0 minecraft:rail[shape=south_west]`

SUCCESS CUE A rail is present at X=6, Y=101, Z=0.

13 **INFERRED**

Open chat and execute the exact command for the southwest corner of the outer loop.

INPUT `/setblock 0 101 6 minecraft:rail[shape=north_east]`

SUCCESS CUE A rail is present at X=0, Y=101, Z=6.

14 **INFERRED**

Open chat and execute the exact command for the southeast corner of the outer loop.

INPUT `/setblock 6 101 6 minecraft:rail[shape=north_west]`

SUCCESS CUE A rail is present at X=6, Y=101, Z=6, completing the outer perimeter of rails.

15 **INFERRED**

Open chat and execute the exact command to place the north straight section of the inner rail loop, directly beside the outer loop.

INPUT `/fill 2 101 1 4 101 1 minecraft:rail[shape=east_west]`

SUCCESS CUE Three rails occupy X=2 through X=4 at Y=101, Z=1.

16 **INFERRED**

Open chat and execute the exact command to place the south straight section of the inner rail loop.

INPUT `/fill 2 101 5 4 101 5 minecraft:rail[shape=east_west]`

SUCCESS CUE Three rails occupy X=2 through X=4 at Y=101, Z=5.

17 **INFERRED**

Open chat and execute the exact command to place the west straight section of the inner rail loop.

INPUT `/fill 1 101 2 1 101 4 minecraft:rail[shape=north_south]`

SUCCESS CUE Three rails occupy X=1 at Y=101, Z=2 through Z=4.

18 **INFERRED**

Open chat and execute the exact command to place the east straight section of the inner rail loop.

INPUT `/fill 5 101 2 5 101 4 minecraft:rail[shape=north_south]`

SUCCESS CUE Three rails occupy X=5 at Y=101, Z=2 through Z=4.

19 **INFERRED**

Open chat and execute the exact command for the northwest corner of the inner loop.

INPUT `/setblock 1 101 1 minecraft:rail[shape=south_east]`

SUCCESS CUE A rail is present at X=1, Y=101, Z=1.

20 **INFERRED**

Open chat and execute the exact command for the northeast corner of the inner loop.

INPUT `/setblock 5 101 1 minecraft:rail[shape=south_west]`

SUCCESS CUE A rail is present at X=5, Y=101, Z=1.

21 **INFERRED**

Open chat and execute the exact command for the southwest corner of the inner loop.

INPUT `/setblock 1 101 5 minecraft:rail[shape=north_east]`

SUCCESS CUE A rail is present at X=1, Y=101, Z=5.

22 **INFERRED**

Open chat and execute the exact command for the southeast corner of the inner loop.

INPUT `/setblock 5 101 5 minecraft:rail[shape=north_west]`

SUCCESS CUE A rail is present at X=5, Y=101, Z=5, and the platform now contains a dense arrangement of neighboring rails.

23 **INFERRED**

Stop placing blocks and visually inspect the source platform. Allow any immediate rail-neighbor updates caused by construction to finish. Do not try to correct individual rail shapes after this point; the settled source arrangement is the state that must be saved.

SUCCESS CUE Every intended rail position contains a rail, every rail has stone support, and no further construction command is being executed.

24 INFERRED

Open chat and execute the exact command to place the source structure block one block below and diagonally outside the source region.

INPUT `/setblock -1 99 -1 minecraft:structure_block`

SUCCESS CUE A structure block is present at X=-1, Y=99, Z=-1.

25 INFERRED

Open chat and execute the exact command to place the destination structure block. Its load offset will place the copied structure at X=20 through X=26, Y=100 through Y=101, Z=0 through Z=6.

INPUT `/setblock 19 99 -1 minecraft:structure_block`

SUCCESS CUE A second structure block is present at X=19, Y=99, Z=-1, and its destination volume is still empty.

26 INFERRED

Open chat and teleport to a position facing the source structure block.

INPUT `/tp @s -1 101 -3 0 30`

SUCCESS CUE The source structure block is visible in front of and below the player.

27 VERIFIED

Use the interaction control, default right-click, on the source structure block to open its interface.

INPUT `right-click`

SUCCESS CUE The Structure Block interface is open.

28 VERIFIED

Use the mode button in the Structure Block interface until the block is in Save mode.

INPUT `Mode: Save`

SUCCESS CUE The interface displays Save mode and exposes structure name, relative position/offset, and size fields.

29 INFERRED

Enter the exact structure name in the Structure Name field. Use the same capitalization and spelling later in the destination block.

INPUT `mc104231_rail_art`

SUCCESS CUE The Structure Name field contains mc104231_rail_art.

30 INFERRED

Set the three Relative Position or Offset fields to X=1, Y=1, Z=1. These values make the saved volume begin at X=0, Y=100, Z=0 relative to the source block at X=-1, Y=99, Z=-1.

INPUT `X=1, Y=1, Z=1`

SUCCESS CUE The save offset fields display 1, 1, and 1 in X, Y, Z order.

31 **INFERRED**

Set the structure Size fields to X=7, Y=2, Z=7. The two Y layers include the stone floor at Y=100 and rails at Y=101.

INPUT `X=7, Y=2, Z=7`

SUCCESS CUE The size fields display 7, 2, and 7 in X, Y, Z order.

32 **VERIFIED**

Set Include Entities to OFF. Enable the bounding-box display if the interface provides that option, then check the rendered outline before saving.

INPUT `Include Entities: OFF; Show Bounding Box: ON`

SUCCESS CUE The bounding box encloses only the 7 by 7 stone platform and its rail layer; it does not include either structure block.

33 **VERIFIED**

Click the Save button once. Wait for the save operation to complete before closing the interface.

INPUT `Save`

SUCCESS CUE The interface or game feedback confirms that the structure named mc104231_rail_art was saved.

34 **VERIFIED**

Close the Structure Block interface with Done or Esc without changing the source rails.

INPUT `Done or Esc`

SUCCESS CUE The game view returns and the settled source rail arrangement remains visible and untouched.

35 **INFERRED**

Open chat and teleport to a position facing the destination structure block.

INPUT `/tp @s 19 101 -3 0 30`

SUCCESS CUE The destination structure block is visible in front of and below the player, with empty space behind it at X=20 through X=26.

36 **VERIFIED**

Use the interaction control, default right-click, on the destination structure block.

INPUT `right-click`

SUCCESS CUE The destination Structure Block interface is open.

37 **VERIFIED**

Use the mode button until the destination structure block is in Load mode.

INPUT `Mode: Load`

SUCCESS CUE The interface displays Load mode and exposes the structure name, relative position/offset, rotation, and mirror controls.

38 **INFERRED**

Enter the exact saved structure name in the Structure Name field.

INPUT

SUCCESS CUE The load interface contains exactly mc104231_rail_art as the requested structure.

39 **INFERRED**

Set the destination Relative Position or Offset fields to X=1, Y=1, Z=1. This selects X=20, Y=100, Z=0 as the copied structure's minimum corner.

INPUT

SUCCESS CUE The load offset fields display 1, 1, and 1 in X, Y, Z order.

40 **REPORT**

Set Rotation to None or 0 degrees and Mirror to None. If Integrity and Seed controls are shown, leave Integrity at 1.0 and leave the Seed field blank. Do not select 90-, 180-, or 270-degree rotation for this primary reproduction.

INPUT

SUCCESS CUE The structure is configured for a complete, unmirrored, unrotated load.

41 **VERIFIED**

Click Load once to prepare the structure and display its target bounding box. This preparation click is not the trigger. Wait until the preview appears and confirm that the target volume is still ready for placement.

INPUT

SUCCESS CUE A 7 by 2 by 7 preview or bounding box appears from X=20 through X=26, Y=100 through Y=101, Z=0 through Z=6, and the interface offers Load again.

42 **REPORT** **TRIGGER**

Click Load a second time to place the saved rail structure for the first time. Do not click Load again after the blocks appear, because the supplied report distinguishes the first load from a subsequent load.

INPUT

SUCCESS CUE The stone floor and rail arrangement have just appeared inside the destination bounding box at X=20 through X=26.

HANDOFF TO OBSERVATION

The saved structure has just been physically placed at the destination by the first actual structure-load operation, using no rotation and no mirror. The source arrangement remains available nearby for comparison. Do not perform a second placement; immediately hand control to the observation subsystem.

UNCERTAINTIES

- The supplied report refers to Step1.png and Step2.png for the exact rail art and required world directions, but the attachment contents are not available. The deterministic adjacent-loop pattern in this plan is therefore a practical inference rather than an exact reconstruction of the reporter's pictured structure.
- Because the exact attachment layout is missing, this substitute pattern may not exercise the same rail-neighbor ordering as the original screenshot. If the first attempt is inconclusive, obtaining the original attachments is preferable to inventing additional patterns.
- Adjacent rails can update one another while the source is being constructed. The procedure intentionally saves the settled source state after all commands have run; the command-requested shape of each individual rail should not be assumed to remain unchanged during construction.
- The report does not specify a seed, coordinates, multiplayer server, or dedicated-server requirement. The Superflat single-player world and fixed coordinates are practical choices intended to remove terrain and exploration variables.
- The Java structure-block Save/Load workflow, including structure names, X/Y/Z size and offset fields, and bounding-box verification, was externally checked against Minecraft Wiki documentation. ([minecraft.wiki] (https://minecraft.wiki/w/Tutorial%3AMoving_structures_using_structure_blocks_from_world_to_world?utm_source=openai))
- Java structure-block interfaces can use an initial Load click to prepare or preview the saved structure before a subsequent click places it. The trigger is specifically the click that physically places blocks. If this exact 1.21 client unexpectedly places blocks on the first click, treat that first placement click as the trigger and hand off immediately rather than clicking again.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Trigger reached?	YES / NO
Crash file	
Notes	
Tested by / date	